Saikiran Chepa

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Professional Summary

Embedded DSP and AI/ML Engineer with 10+ years of experience delivering measurable performance gains across the full engineering stack — from bare-metal assembly on ARM, TI DSP, and Qualcomm Hexagon platforms to training and deploying deep learning models on edge devices. Adept at bridging low-level systems engineering (fixed-point arithmetic, SIMD intrinsics, Linux system programming) with modern AI practices (LLMs, RAG, RLHF), with a track record of reducing compute cycles by up to 60% and shipping production-ready AI systems end-to-end. GATE DA 2026 qualified; NPTEL Domain Scholar in Signal Processing & Communications; recognized for clear technical communication, cross-functional collaboration, and rapid adaptation to new architectures, toolchains, and project requirements.

Key Highlights

- **Languages & Assembly:** C, C++, Python, MATLAB, and bare-metal assembly — ARM NEON intrinsics, TI Linear Assembly (C64x+/C66x), Qualcomm QDSP intrinsics.
- **Embedded Protocols:** Hands-on with SPI, I2C, I2S (ALSA/DAC audio serial data path), and UART — register-level configuration and on-target debugging.
- **Linux & ALSA:** Built a multi-threaded real-time ALSA audio engine — pthreads producer–consumer (mutex + condvar), XRUN detection/recovery, `SCHED_FIFO` real-time scheduling, and adaptive resampling for VoIP clock-drift cancellation.
- **Linux System Programming:** POSIX threads, IPC (named pipes, shared memory, message queues, sockets), character device drivers, kernel modules (`insmod/rmmod`), `mmap`, `fork/exec`, and kernel builds from source.
- **DSP Performance:** Reduced compute cycles by ~60% via float-based loop optimizations in a commercial CarVoice pipeline; closed ARMv8-A vs. ARMv7-A gap from 2.2× to 1.4× using 25 hand-optimized ARM NEON functions.
- **DSP Algorithms & Codecs:** FIR/IIR filters, FFT/IFFT, multirate DSP, fixed-point arithmetic; optimized SILK, Speex, Opus, eAAC+, iLBC, and AGVC codecs on DSP and SIMD architectures.
- **Processor Architectures:** ARMv7-A, ARMv8-A/ARM64, Qualcomm Hexagon DSP, TI C64x+/C66x, SHARC DSP, Intel AVX2, RISC-V DSP.
- **Edge AI Deployment:** Trained ResNet18 on CIFAR-10 (~90% top-1 accuracy); deployed on Qualcomm Hexagon DSP, GPU, and CPU via SNPE pipeline with Android NDK cross-compilation and C++ dynamic loader (`dlopen/dlsym`).
- **AI & LLMs:** LLM fine-tuning (SFT, RLHF, PPO, DPO, GRPO), quantization (Bitsandbytes, AWQ, SmoothQuant, QAT), RAG pipelines, multi-modal AI; shipped a production AWS system using Gemini LLM, WebSockets, and MCP.
- **Audio & Voice Systems:** End-to-end audio processing expertise spanning commercial CarVoice and VoiceExperience pipelines, ALSA-based Linux audio engines, MATLAB-based simulation with CSIM API development for software-in-the-loop validation, FFT/IFFT integration (RISC-V, SHARC), and real-time audio scheduling on embedded Linux.
- **Cloud & Cross-Platform Deployment:** AWS EC2, SageMaker, and S3 for hosting and serving AI backends; Android NDK cross-compilation for ARM64 targets; Raspberry Pi 4 for embedded benchmarking; experience shipping models on both cloud infrastructure and edge hardware (Snapdragon, Hexagon DSP) in the same end-to-end pipeline.
- **Development Tools & Environments:** CMake, Makefile, Git, SVN, Python/shell/batch scripting, ADB; IDEs: CCS (Code Composer Studio), Eclipse, DS-5, VS Code, Cursor AI; Langchain, LangGraph, MCP protocol, and OpenAI SDK for AI agent and tool-orchestration workflows.
- **Certifications & Academia:** GATE DA 2026 Qualified; NPTEL Domain Scholar in Signal Processing & Communications (60 weeks, IIT-level); Deep Learning & Math for ML Specializations (Coursera); ERA V3 — The School of AI, Bangalore.

Professional Experience

Consultant Senior Staff Engineer **Zentree Labs, Hyderabad**
Jan 2026 – May 2026

- Trained ResNet18 CNN on CIFAR-10 using PyTorch Lightning; achieved ˜90% top-1 accuracy with systematic hyperparameter tuning.
- Built full deployment pipeline: PyTorch → ONNX → SNPE (Snapdragon Neural Processing Engine) compilation (`snpe-onnx-to-dlc`).
- Generated shared libraries for DSP (`snpe-dsp-lib-generator`), GPU, and CPU backends; automated backend evaluation and accuracy comparison.
- Deployed model on Snapdragon board across CPU, GPU, and Hexagon DSP.
- Cross-compiled ARM64 host application using Android NDK; developed C++ dynamic loader via `dlopen`/`dlsym`.
- Familiar with LLM quantization using Bitsandbytes; AWQ, SmoothQuant, and QAT.

Engineer III **Capgemini (Client: Qualcomm), Hyderabad**
Dec 2016 – May 2023 — Jan 2025 – Mar 2025

- Converted floating-point speech algorithms to fixed-point; implemented `log10` and `pow` using fixed-point primitives `log2`/`pow2`.
- Used MATLAB Codegen to produce C code; refactored scripts to resolve codegen errors and replaced dynamic arrays with fixed-size implementations for embedded constraints.
- Developed CSIM APIs for MATLAB-based DSP models, enabling software-in-the-loop validation.
- Optimized stereo FIR filter with decimation-by-2 in assembly: skipped alternate samples and computed both channels simultaneously to maximize instruction slot utilization.
- Implemented upsampling followed by FIR filtering for multirate DSP pipelines.
- Automated build, compile, and test pipelines using Python, batch/shell scripts, and CMake.
- Optimized matrix multiplication for AGVC (AI/ML-based speech codec) targeting Intel AVX2.

Senior DSP Engineer **Capgemini (Client: Goodix), Bangalore**
Aug 2023 – Jul 2024

- Integrated RISC-V DSP library FFT/IFFT into VoiceExperience; validated non-bit-exact outputs using MATLAB RMSE analysis (dB) and Adobe Audition.
- Bridged SHARC floating-point FFT/IFFT into CarVoice's fixed-point pipeline via float↔fixed conversions; optimized inner loops by processing in floating-point then converting back, achieving ˜60% cycle reduction.
- Closed ARMv8-A vs. ARMv7-A performance gap from 2.2× to 1.4× by hand-coding 25 hot functions with ARM NEON intrinsics.
- Developed Python scripts to compare ARMv7-A and ARMv8-A flat profiles and identify un-optimized functions.
- Measured MCPS on Android via ADB; profiled at function level using Simpleperf.
- Automated MCPS benchmarking on Raspberry Pi 4: minimum of 5 runs per frame, average and peak MCPS across the full dataset.

Senior DSP Engineer **Couth Infotech, Hyderabad**
Apr 2012 – Mar 2015

- Optimized SILK, Speex, and Opus codecs for TI C66x and C64x+ DSPs using linear assembly.
- Partitioned scratch and channel buffers into 4 KB blocks; reduced stack usage through scratch-based memory management.
- Set up Network Development Kits (NDK) for C6678 and C6472 boards; validated codec against standard and non-standard test vectors.
- Authored compliance wrappers: DataMove, illegal read/write detection, register preservation, buffer corruption detection, I/O alignment, interrupt testing, stack sizing, code coverage, and scratch contamination checks.

Senior Software Engineer **Aricent, Hyderabad**
Jul 2005 – Jul 2009

- Optimized eAAC+ decoder for TI C64x+ DSP; developed interleaved and non-interleaved output modes.
- Ported iLBC codec from 32-bit floating-point to fixed-point representation.
- Optimized audio and speech codecs across TI C64x+ and C66x DSP platforms.

Projects

Multi-Threaded Real-Time ALSA Audio Engine *Zentree Labs, Jan 2026 – May 2026*

- Applied ALSA's 3-layer architecture: user-space PCM writes via `snd_pcm_writei()` into the kernel ring buffer → autonomous DMA transfer to hardware FIFO → I^2S delivery to DAC, with zero CPU involvement in the data path once DMA is armed.
- Implemented producer–consumer threading with POSIX pthreads: network/decoder thread fills a thread-safe PCM queue (mutex + condition variable); ALSA playback thread dequeues one period per wakeup, deliberately minimizing work per period to avoid scheduling jitter.
- Implemented XRUN (underrun/overrun) detection via `hw_ptr/appl_ptr` divergence monitoring; recovery via `snd_pcm_prepare()` on `-EPIPE`; used `snd_pcm_wait()` for efficient period-aligned blocking to eliminate busy-polling.
- Applied `SCHED_FIFO` (priority 80) real-time scheduling to the playback thread to minimize CPU scheduling jitter and prevent underruns under system load.
- Implemented jitter buffer and adaptive resampling for VoIP clock-drift compensation: monitored buffer fill level as a PLL error signal and adjusted effective playback ratio (±0.1%) to continuously cancel cross-device hardware clock mismatch (~78 samples/min drift at 48 kHz).

Linux System Programming

Hands-on experience with core Linux and POSIX programming concepts essential for embedded, systems, and driver development:

- **Multithreading & Synchronization:** POSIX threads (`pthreads`), thread lifecycle management, mutexes, condition variables, semaphores, and deadlock avoidance strategies.
- **Memory & Process Management:** Virtual memory, `mmap`, dynamic memory allocation, process creation (`fork/exec`), zombie/orphan handling, and signals.
- **IPC Mechanisms:** Named pipes (FIFOs), POSIX shared memory, message queues, and POSIX/BSD sockets for inter-process communication.
- **Device Drivers:** Character device driver development, kernel module programming (`insmod/rmmod`), `/proc` and `/sys` filesystem interfaces.
- **Build Infrastructure:** GNU toolchain (`gcc`, `ld`, `objdump`, `nm`, `readelf`), Makefile authoring, cross-compilation toolchains, and building the Linux kernel from source (`menuconfig`, modules, device tree overlays).

Certifications & Achievements

GATE DA (Data Science & AI) 2026 Qualified

Domain Scholar – Signal Processing & Communications NPTEL, 60 Weeks

- Discrete Time Signal Processing, Multirate DSP, Adaptive Signal Processing
- Applied Linear Algebra, Probability Foundations for Electrical Engineers, Introduction to Information Theory, Principles of Signals & Systems

Mathematics for Machine Learning and Data Science Specialization Coursera

- Linear Algebra, Calculus, Probability & Statistics for ML
- Key skills: Bayesian Statistics, Dimensionality Reduction, Sampling, Numerical Analysis, NumPy

Deep Learning Specialization Coursera

- Designed and trained CNNs, RNNs, and transformer-based models using TensorFlow
- Applied hyperparameter tuning, bias-variance analysis, object detection, neural style transfer

ERA V3 (Extensive Reimagined AI, v3) The School of AI, Bangalore

- PyTorch hands-on: neural networks, CNNs, ResNets, and optimization techniques
- Trained transformer models from scratch; studied LLM fine-tuning strategies: SFT, RLHF, PPO, DPO, GRPO
- Multi-modal AI, AI agents, Model Context Protocol (MCP), Retrieval-Augmented Generation (RAG)
- **Capstone: Intelligent Email Query Assistant**
  - Built client-server application enabling natural-language querying of Gmail
  - Gmail API OAuth authentication (client-side); Gemini LLM backend on AWS
  - Regex-based filtering; server-side transaction extraction; MCP-based calculator orchestration
  - Real-time chat interface using WebSockets

Education

MTech in Electrical Engineering IIT Madras, 2005
BTech in Electronics & Communication Engineering GRIET, JNTU Hyderabad, 2003

Technical Skills

- **Programming Languages:** C, C++, Python, MATLAB, Assembly (ARM NEON intrinsics, TI Linear Assembly, Qualcomm QDSP intrinsics)
- **Embedded & DSP:** Fixed-point arithmetic, FIR/IIR filters, FFT/IFFT, multirate DSP, floating-to-fixed conversion, CSIM API development, codec optimization (SILK, Speex, Opus, eAAC+, iLBC, AGVC)
- **Embedded Protocols & Interfaces:** SPI, I^2C, I^2S (audio serial interface — integral to ALSA/DAC data path), UART; familiar with protocol timing, register-level configuration, and on-target debugging
- **SIMD & Performance Optimization:** ARM NEON intrinsics, Intel AVX2, MCPS profiling, cycle reduction, Simpleperf, ADB-based benchmarking
- **Processor Architectures:** ARMv7-A, ARMv8-A/ARM64, Qualcomm Hexagon DSP, Intel AVX2, RISC-V DSP, TI C64x+, TI C66x, SHARC DSP
- **Linux System Programming:** POSIX threads, mutexes, semaphores, shared memory, named pipes, sockets, character device drivers, kernel modules, GNU toolchain, kernel build from source, ALSA (`libasound`)
- **AI & Deep Learning:** CNNs (ResNet18), RNNs, Transformers, LLMs, RAG, multi-modal AI, edge AI deployment, fine-tuning (SFT, RLHF, PPO, DPO, GRPO), LLM quantization (Bitsandbytes, AWQ, SmoothQuant, QAT)
- **Frameworks & Libraries:** PyTorch, PyTorch Lightning, ONNX, Qualcomm QNN SDK, Qualcomm SNPE, MATLAB Codegen, Hugging Face Transformers, TensorFlow
- **Cloud & Deployment:** AWS EC2, AWS SageMaker, AWS S3, cross-compilation (ARM64, Android NDK), dynamic linking (`dlopen/dlsym`)
- **Build & Automation:** CMake, Makefile, Git, SVN, Python scripting, shell/batch scripting, ADB
- **Tools:** VS Code, Cursor AI, Claude Code, CCS (Code Composer Studio), Eclipse, DS-5, Langchain, LangGraph, MCP, OpenAI SDK
- **Environments:** Windows, Linux, Android, Raspberry Pi